**Universidad Autónoma de Nuevo León**

**Preparatoria 7 “Dr. Óscar Vela Cantú”**

# Chemical Phenomena in the Environment

## Learning Evidence 3: STEM


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## Introduction

On January 2 of 2022, the government emitted an emergency declaration for drought in Nuevo León, reducing water supply up to 6 hours a day (Gobierno Constitucional del Estado Libre y Soberano de Nuevo León, 2022). This drastically limited our daily activities or necessities. Aside from the climatic effect “La Niña,” the drought was mostly caused by the poor infrastructure and water management in the city. In the Metropolitan Area, the average water usage in a day is 141 liters per person (lps) (Servicios de Agua y Drenaje, 2023), while the World Health Organization suggests that it should not overpass 100 lps.

Smart Flow is a mobile application developed by our team to help users monitor their water consumption in real time: daily, weekly, monthly, and annually. It connects to smart sensors that measure water flow and immediately alerts users when irregular patterns or potential leaks are detected. Through its intuitive interface, the app raises awareness about personal water usage and promotes responsible consumption habits. Smart Flow aims to make people more conscious of the amount of water they use and encourage sustainable actions that protect this vital resource.

We will measure the effectiveness of our project in the consumer’s life throughout a survey, which will give us specific information about the person and their water consumption before and after the use of the project (sensor and app) for some time.

**Development**

The sensors installed in the water network continuously collect data on flow and pressure, transmitting it securely to the application. The app analyzes this information and presents it through detailed charts, graphs, and statistics, allowing users to understand their daily, weekly, monthly, and annual water consumption patterns with precision. When Smart Flow detects unusual spikes in water usage or leaks, it instantly sends alerts to notify the user, helping to prevent waste and unexpected high bills.

The application also allows users to set personal goals, compare their progress over time, and receive personalized recommendations to reduce their water footprint. By combining technology and environmental awareness, Smart Flow not only tracks water usage but also educates and empowers users to act toward a more sustainable lifestyle.

The mathematical procedures that were necessary to create the prototype and analyze, measuring, and helping to manage the water usage is a system that calculates the real-time water flow rate using pulse signals from the flow sensor connected to the ESP32, which is a Bluetooth microcontroller that provides control to certain devices. Each pulse represents a specific water volume, and we can calculate it with the equation:

$$\textbf{Flow Rate (L/min)} = \frac{\textbf{Pulses per second x 60}}{\textbf{450}}$$

The total water consumption is calculated by the equation:

$$\textbf{Total volume (L)} = \Sigma\ \textbf{(Flow Rate x Time Interval)}$$

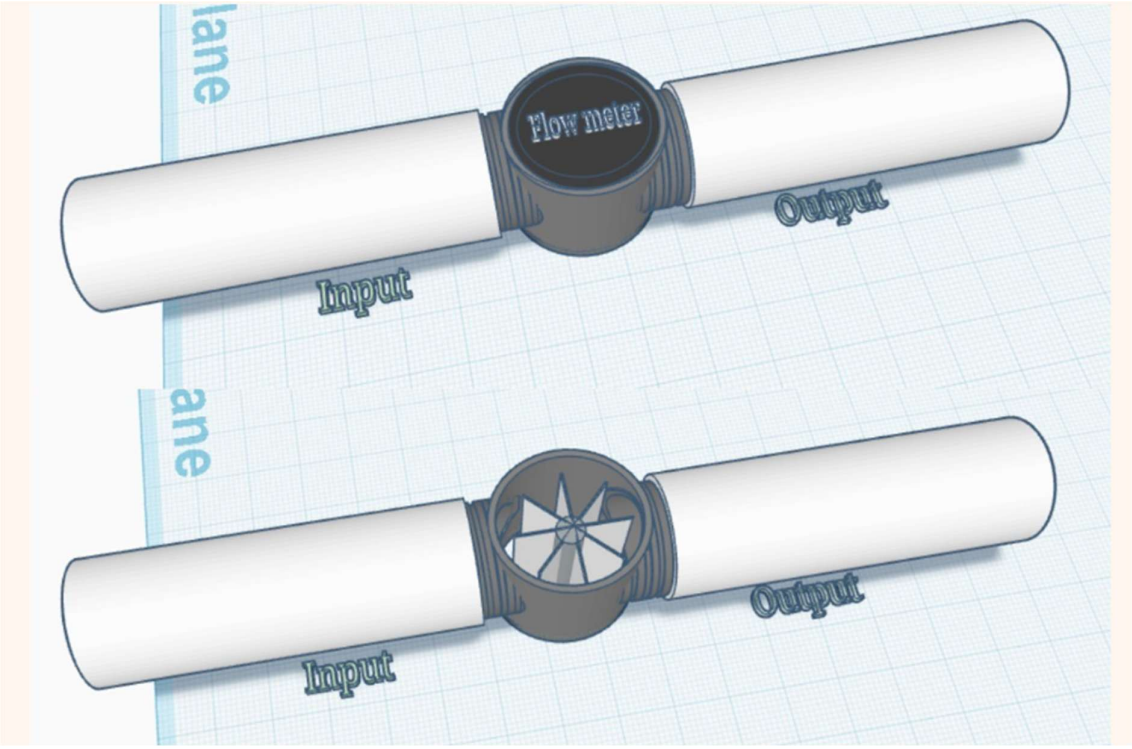


*Figure 1. Tinkercad prototype with and without the sensor cap.*

The Hall Effect Sensor flow meter, integrated with a turbine rotor, is presented as an optimal solution primarily because it achieves high accuracy and durability while remaining low cost (Sood, Kaur, & Lenka, 2013). This replaces high-cost alternatives which would not benefit the user economically at all.

In contrast, the mobile app is designed to be attractive and to generate awareness through milestones and goals, also known as “gamification.” As an example, the language learning app Duolingo uses a similar method: it employs lingots (jewel symbol used to reward students who complete activities and levels), which can be redeemed for additional practices. Another strategy is the “Daily Goal” feature that rewards persistence and helps students re-engage in work by offering stunning sounds and unique graphics. (Sari, Hadina, & Rahmanda, 2022)

Finally, the Arduino platform is strongly recommended for its open-source software and hardware, cost-effective components, and language based on C++ which features a simple and straightforward syntax that makes microcontroller programming easy to learn for beginners and facilitates applications in diverse areas like robotics or, in this case, home automation. (Chatterjee, 2023)

The variables that we considered while creating this prototype are surveys, water pipes, sensors, and water itself. Surveys are a main component of this project because with their help, we will be able to find out whether our project is useful and helpful to people. The water pipes help distribute water into the houses and regulate its flow. Sensors detect how much water passes through the pipe, and with the water, we will only count how many liters flow inside the pipes into the houses. The pipe connections measure ½ inches, and the sensor measures 62.5 mm long, 36 mm wide, and 35 mm high.

The chemical formula that we are using is:

$$4Fe(s) + 3O_2(g) + 2H_2O(aq) \rightarrow 2Fe_2O_3H_2O(s)$$

Overuse of water can cause corrosion in the drainage and/or water entry pipes; we will use the corrosion chemical equation as a method to confirm our project is working.

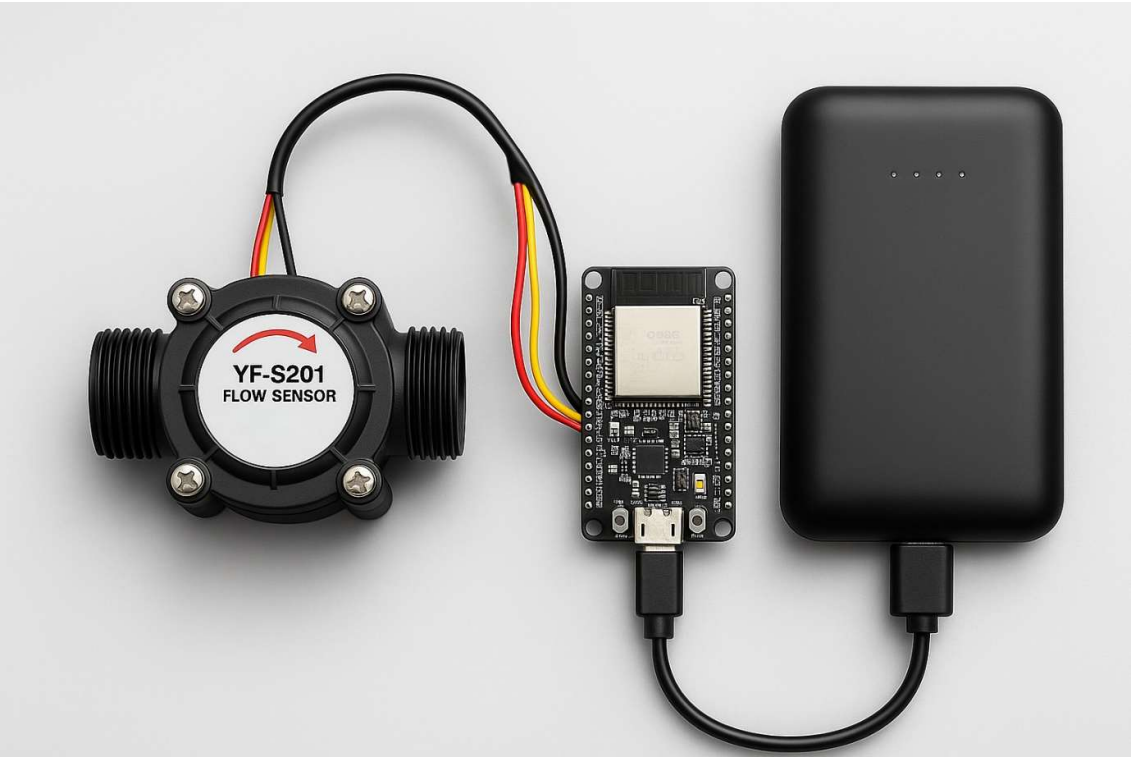


*Figure 2. Basic configuration of Flow Sensor YF-S201 with ESP32 and mobile power bank. (Enhanced by AI)*

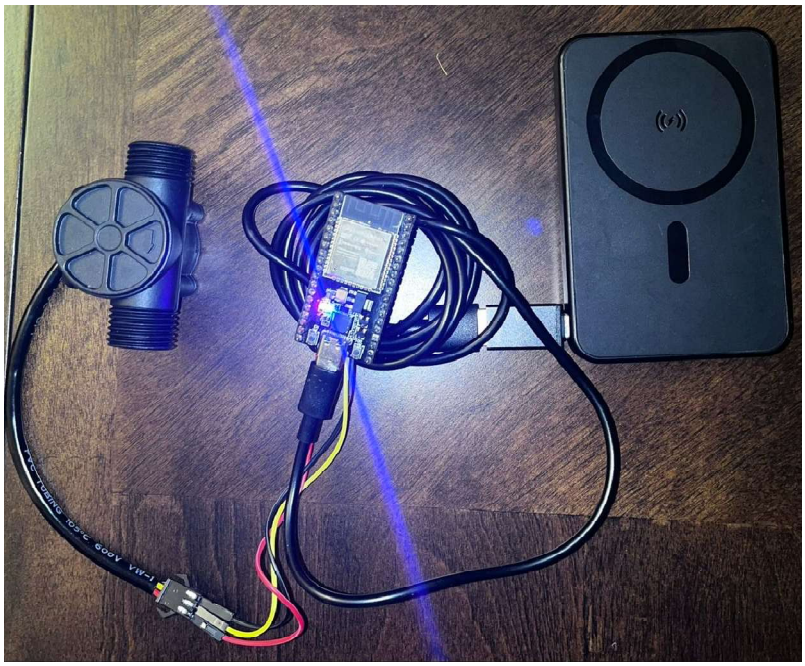

*Figure 3. Physical assembly of the project, showing interconnected YF-S201 Flow Sensor, Esp32 microcontroller, and power bank.*

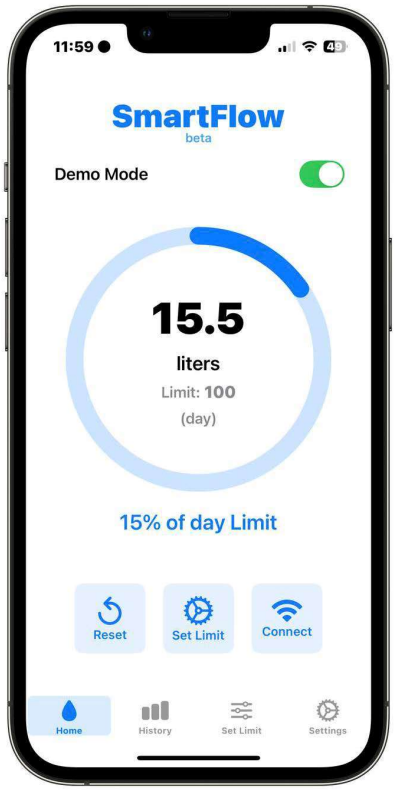


*Figure 42. Screenshot of the mobile application "Smart Flow" in function.*

The Smart Flow mobile application offers important benefits for the users and the environment. It helps people become more aware of their daily water consumption by displaying real-time data collected from a water meter connected to the app. This allows users to identify wasteful habits and set personal goals to reduce water use. By turning water monitoring into a fun and easy activity, Smart Flow promotes environmental responsibility and encourages sustainable behavior in households. The app contributes to water conservation and supports the responsible use of natural resources.

## Conclusion

The Smart Flow project was developed to address the growing problem of water scarcity, especially the daily water issues faced in Nuevo León. The region has experienced frequent shortages and uneven water distribution; we need greater awareness about efficient consumption. Smart Flow, integrating hardware sensors with an easy interface, brings accessibility and positive environmental impact. By turning water tracking into an easy and motivating experience, the project helps people to change their daily habits and encourages responsible use of resources. The combination of real-time data and goal settings like Duolingo®. Using the interactive part to make the people of Nuevo León and beyond, being conscious about environmental responsibility in a fun and educational way.

Although, even if Smart flow presents a feasible and environmentally beneficial proposal, its full viability cannot yet be assured since the experimentation requires, at least, one month of continuous use in different households to gather accurate data and evaluate performance. Despite this limitation, the project demonstrates strong potential to promote responsible water use and awareness. Once evaluated over time, Smart Flow could prove to be an effective and sustainable tool for reducing water waste and encouraging long-term behavioral change in Nuevo León and beyond.

## References


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**Link to survey:** https://forms.office.com/r/XHTtQhq4Z0

**Link to GitHub Repository:** https://github.com/r0bledas/SmartFlow